Data Analysis
AMS 315

Chap 4: Basic Probability

Learning Objectives

In this chapter, you learn:

- Basic probability concepts and definitions
- Conditional probability
- To use Bayes' Theorem to revise probabilities
- Various counting rules

Important Terms

- Probability – the chance that an uncertain event will occur (always between 0 and 1)
- Event – Each possible outcome of a variable
- Simple Event – an event that can be described by a single characteristic
- Sample Space – the collection of all possible events

Assessing Probability

- There are three approaches to assessing the probability of an uncertain event:

1. *a priori* classical probability

$$\text{probability of occurrence} = \frac{X}{T} = \frac{\text{number of ways the event can occur}}{\text{total number of elementary outcomes}}$$

2. empirical classical probability

$$\text{probability of occurrence} = \frac{\text{number of favorable outcomes observed}}{\text{total number of outcomes observed}}$$

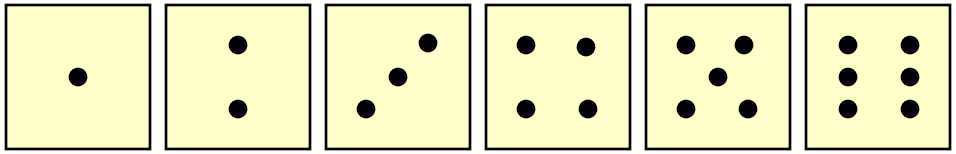
3. subjective probability

an individual judgment or opinion about the probability of occurrence

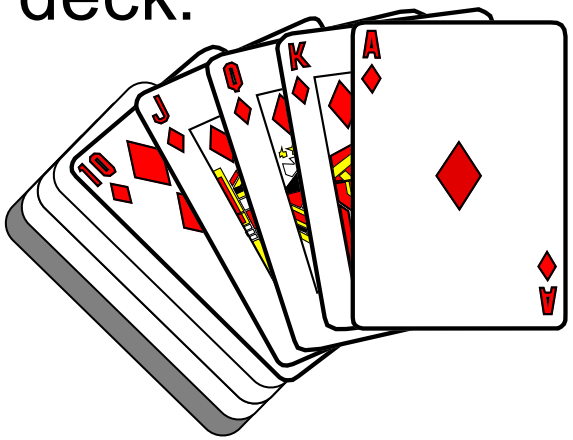
Sample Space

The Sample Space is the collection of all possible events

e.g. All 6 faces of a die:



e.g. All 52 cards of a bridge deck:



Events

- Simple event
  - An outcome from a sample space with one characteristic
  - e.g., A red card from a deck of cards
- Complement of an event A (denoted A')
  - All outcomes that are not part of event A
  - e.g., All cards that are not diamonds
- Joint event
  - Involves two or more characteristics simultaneously
  - e.g., An ace that is also red from a deck of cards

Visualizing Events

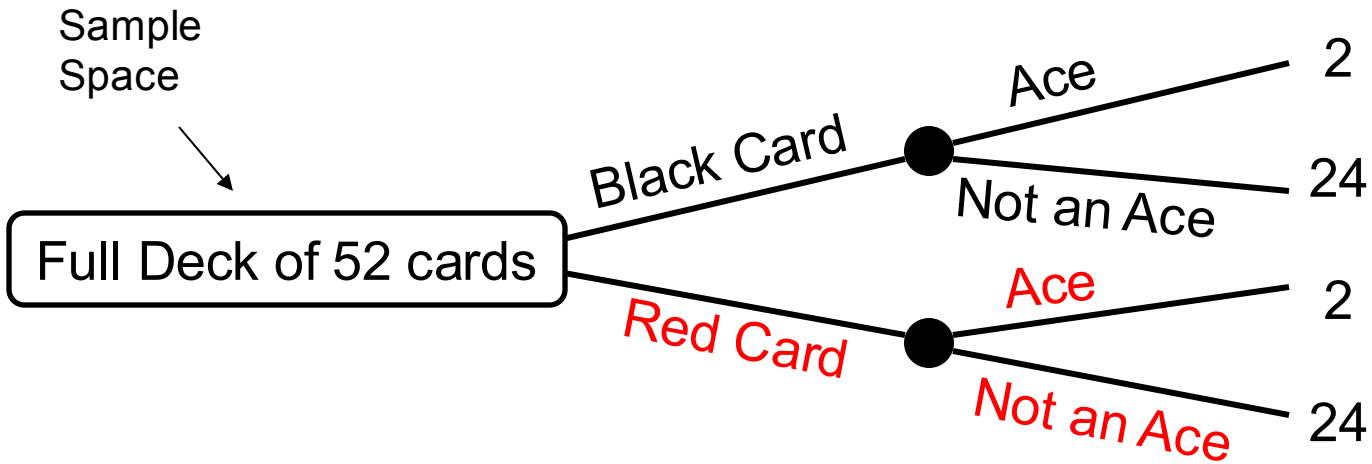
- Contingency Tables

	Ace	Not Ace	Total
Black	2	24	26
Red	2	24	26
Total	4	48	52

Sample Space

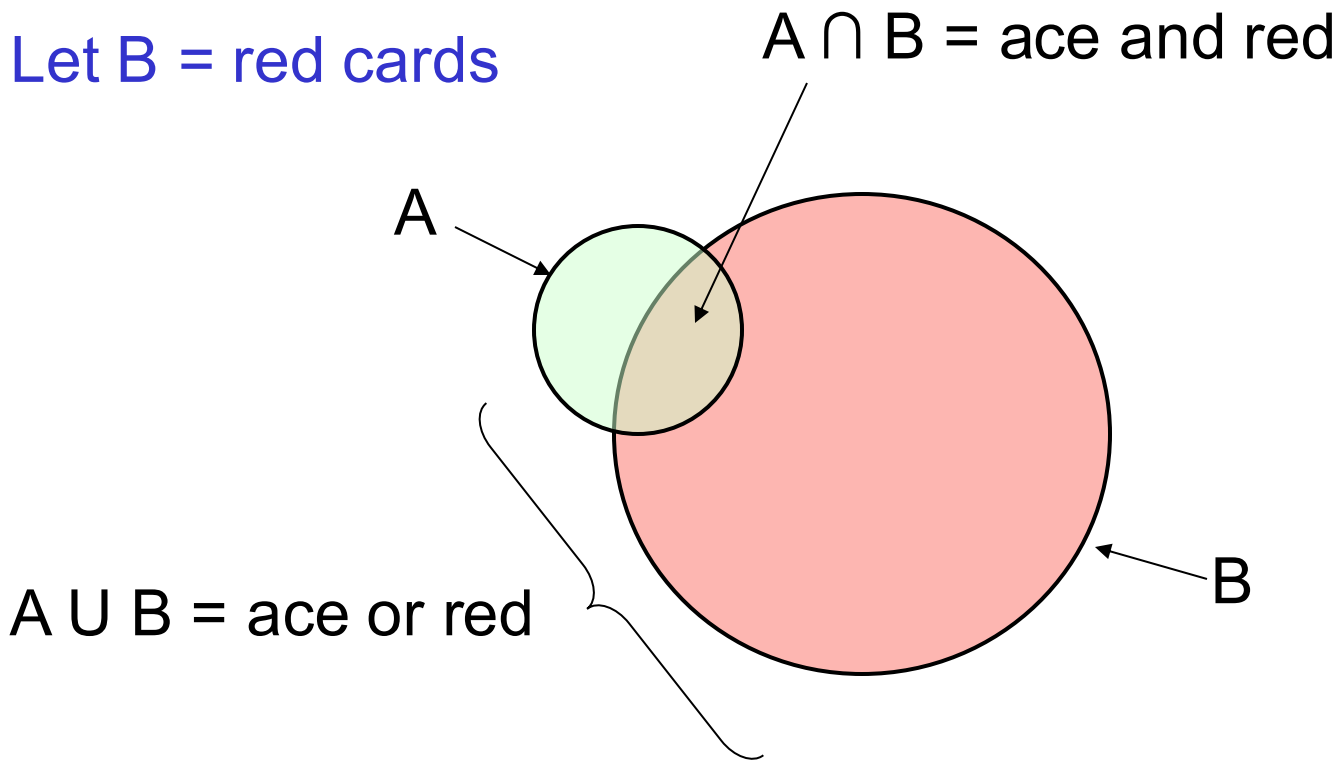
Visualizing Events

- Tree Diagrams



Visualizing Events

- Venn Diagrams
  - Let A = aces
  - Let B = red cards



Mutually Exclusive Events

- Mutually exclusive events
  - Events that cannot occur together

example:

A = queen of diamonds; B = queen of clubs

  - Events A and B are mutually exclusive

Collectively Exhaustive Events

- Collectively exhaustive events
  - One of the events must occur
  - The set of events covers the entire sample space

Example:

A = aces; B = black cards;
C = diamonds; D = hearts

  - Events A, B, C and D are collectively exhaustive (but not mutually exclusive – an ace may also be a heart)
  - Events B, C and D are collectively exhaustive and also mutually exclusive

Probability

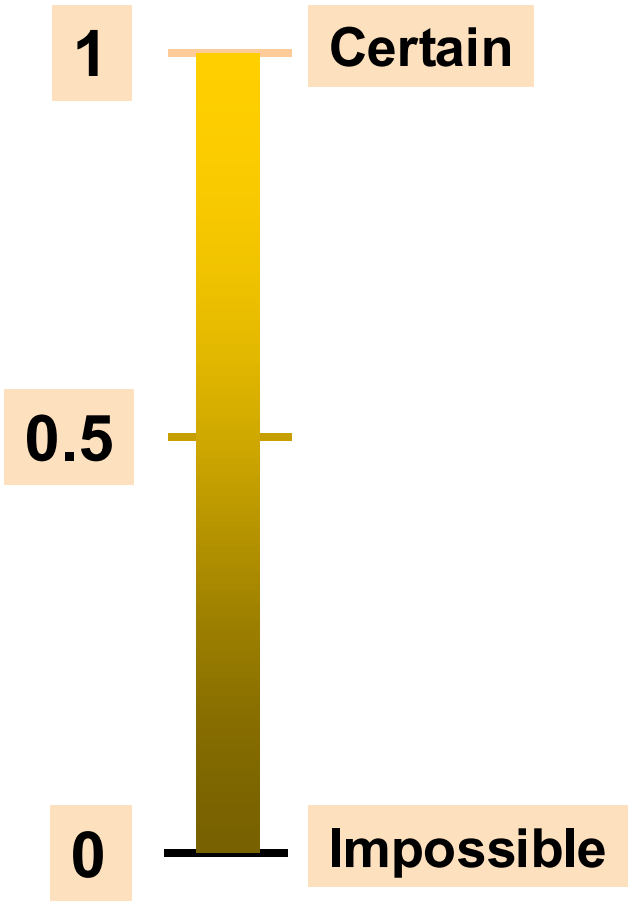
- Probability is the numerical measure of the likelihood that an event will occur
- The probability of any event must be between 0 and 1, inclusively

$$0 \le P(A) \le 1 \quad \text{For any event A}$$

- The sum of the probabilities of all mutually exclusive and collectively exhaustive events is 1

$$P(A) + P(B) + P(C) = 1$$

If A, B, and C are mutually exclusive and collectively exhaustive



Computing Joint and Marginal Probabilities

- The probability of a joint event, A and B:

$$P(A \text{ and } B) = \frac{\text{number of outcomes satisfying A and B}}{\text{total number of elementary outcomes}}$$

- Computing a marginal (or simple) probability:

$$P(A) = P(A \text{ and } B_1) + P(A \text{ and } B_2) + \cdots + P(A \text{ and } B_k)$$

  - Where B_1, B_2, …, B_k are k mutually exclusive and collectively exhaustive events

Joint Probability Example

P(Red and Ace)

$$= \frac{\text{number of cards that are red and ace}}{\text{total number of cards}} = \frac{2}{52}$$

Type	Color		Total
	Red	Black	
Ace	2	2	4
Non-Ace	24	24	48
Total	26	26	52

Marginal Probability Example

P(Ace)

$$= P(\text{Ace and Red}) + P(\text{Ace and Black}) = \frac{2}{52} + \frac{2}{52} = \frac{4}{52}$$

Type	Color		Total
	Red	Black	
Ace	2	2	4
Non-Ace	24	24	48
Total	26	26	52

Joint Probabilities Using Contingency Table

Event	Event		Total
	B_1	B_2	
A_1	$P(A_1 \text{ and } B_1)$	$P(A_1 \text{ and } B_2)$	$P(A_1)$
A_2	$P(A_2 \text{ and } B_1)$	$P(A_2 \text{ and } B_2)$	$P(A_2)$
Total	$P(B_1)$	$P(B_2)$	1

Joint Probabilities

Marginal (Simple) Probabilities

General Addition Rule

General Addition Rule:

P(A or B) = P(A) + P(B) - P(A and B)

If A and B are mutually exclusive, then

P(A and B) = 0, so the rule can be simplified:

P(A or B) = P(A) + P(B)

For mutually exclusive events A and B

General Addition Rule Example

P(Red or Ace) = P(Red) +P(Ace) - P(Red and Ace)

= 26/52 + 4/52 - 2/52 = 28/52

Don't count the two red aces twice!

Type	Color		Total
	Red	Black	
Ace	2	2	4
Non-Ace	24	24	48
Total	26	26	52

Computing Conditional Probabilities

- A conditional probability is the probability of one event, given that another event has occurred:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

The conditional probability of A given that B has occurred

$$P(B \mid A) = \frac{P(A \text{ and } B)}{P(A)}$$

The conditional probability of B given that A has occurred

Where P(A and B) = joint probability of A and B

P(A) = marginal probability of A

P(B) = marginal probability of B

Conditional Probability Example

- Of the cars on a used car lot, 70% have air conditioning (AC) and 40% have a CD player (CD). 20% of the cars have both.

- What is the probability that a car has a CD player, given that it has AC ?

i.e., we want to find $P(CD \mid AC)$

Conditional Probability Example

(continued)

- Of the cars on a used car lot, **70%** have air conditioning (AC) and **40%** have a CD player (CD). **20%** of the cars have both.

	CD	No CD	Total
AC	0.2	0.5	0.7
No AC	0.2	0.1	0.3
Total	0.4	0.6	1.0

$$P(CD \mid AC) = \frac{P(CD \text{ and } AC)}{P(AC)} = \frac{0.2}{0.7} = 0.2857$$

Conditional Probability Example

(continued)

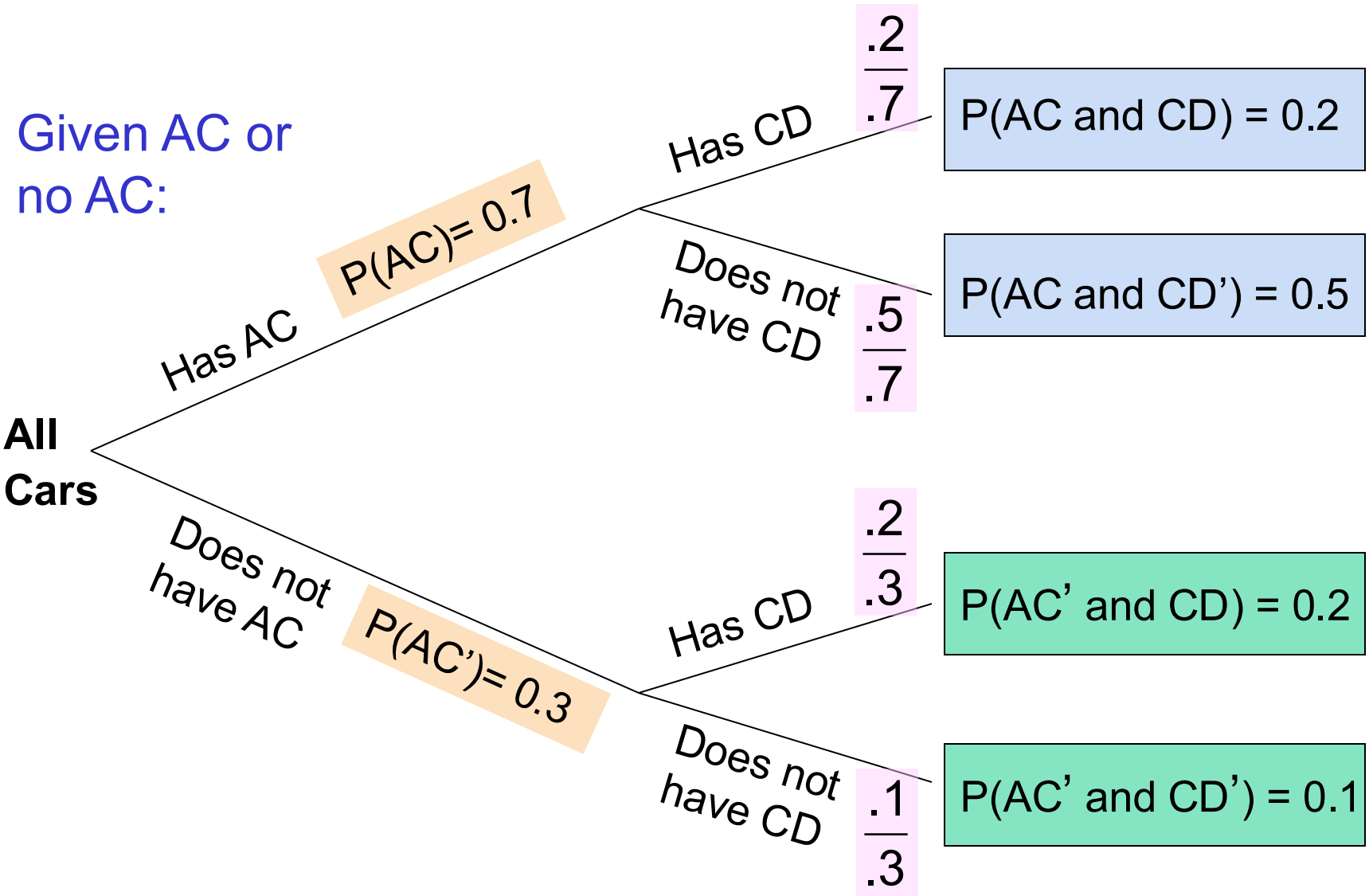
- Given AC, we only consider the top row (70% of the cars). Of these, 20% have a CD player. 20% of 70% is about 28.57%.

	CD	No CD	Total
AC	**0.2**	**0.5**	**0.7**
No AC	**0.2**	**0.1**	**0.3**
Total	**0.4**	**0.6**	**1.0**

$$P(CD \mid AC) = \frac{P(CD \text{ and } AC)}{P(AC)} = \frac{0.2}{0.7} = 0.2857$$

Using Decision Trees

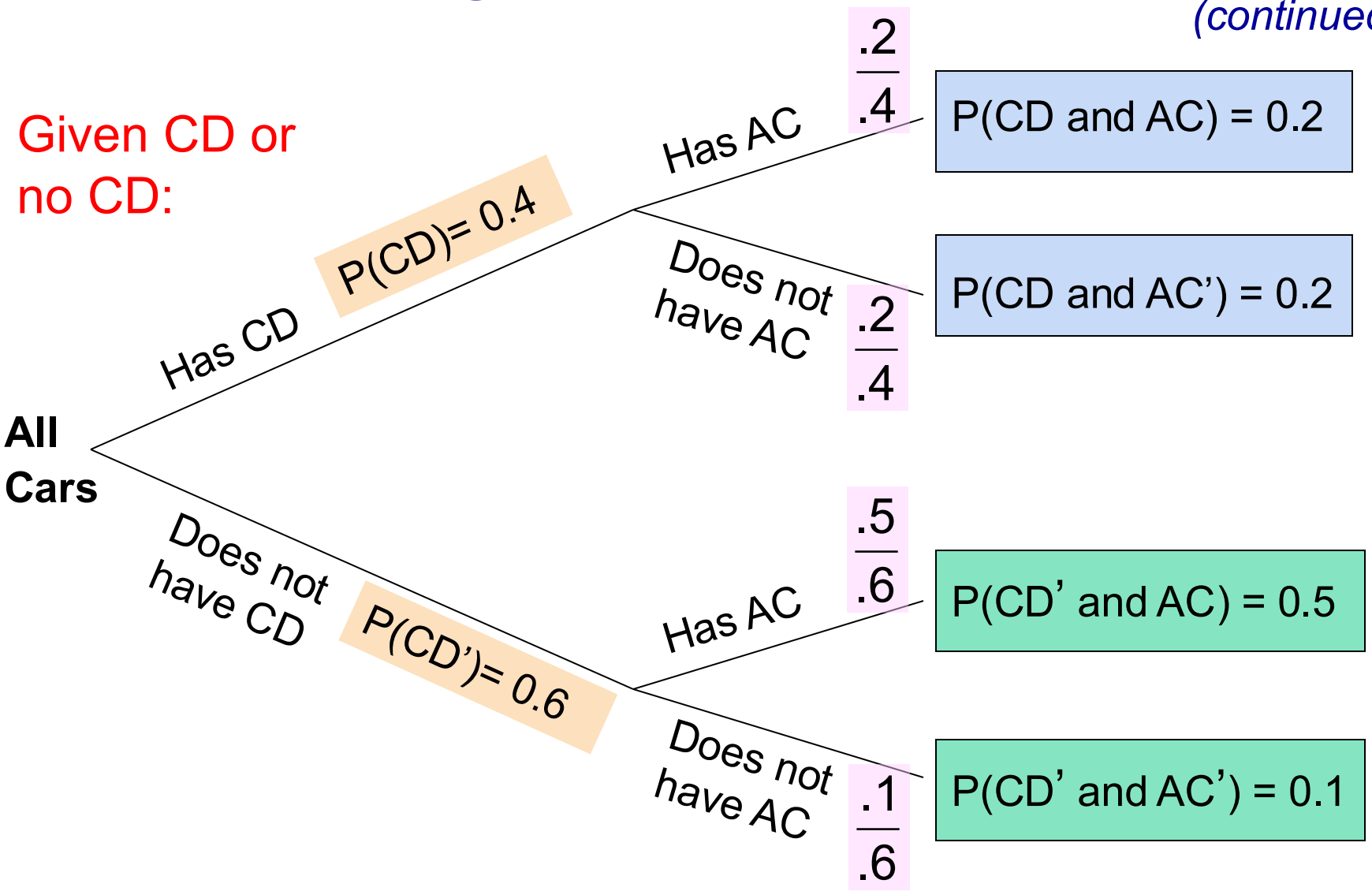
Given AC or no AC:



Using Decision Trees

(continued)

Given CD or no CD:



Statistical Independence

- Two events are independent if and only if:

$$\boxed{P(A \mid B) = P(A)}$$

- Events A and B are independent when the probability of one event is not affected by the other event

Multiplication Rules

- Multiplication rule for two events A and B:

$$\boxed{P(A \text{ and } B) = P(A \mid B)P(B)}$$

Note: If A and B are independent, then $P(A \mid B) = P(A)$ and the multiplication rule simplifies to

$$P(A \text{ and } B) = P(A)P(B)$$

Marginal Probability

- Marginal probability for event A:

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \cdots + P(A \mid B_k)P(B_k)$$

  - Where B_1, B_2, …, B_k are k mutually exclusive and collectively exhaustive events

Bayes' Theorem

- Bayes' Theorem is used to revise previously calculated probabilities when you have new information.
- It was developed by Rev. Thomas Bayes in the eighteenth century.
- Bayes' Theorem is an extension of what you previously learned about conditional probability.

Bayes' Theorem

- Bayes' Theorem applies the following equation:

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \ldots + P(A \mid B_k)P(B_k)}$$

- where:

  B_i = i^{th} event of k mutually exclusive and collectively exhaustive events

  A = new event that might impact $P(B_i)$

Bayes' Theorem Example

- A drilling company has estimated a 40% chance of striking oil for their new well.
- A detailed test has been scheduled for more information. Historically, 60% of successful wells have had detailed tests, and 20% of unsuccessful wells have had detailed tests.
- Given that this well has been scheduled for a detailed test, what is the probability that the well will be successful?

Bayes' Theorem Example

(continued)

- Let S = successful well

  U = unsuccessful well
- $P(S) = 0.4$, $P(U) = 0.6$ (prior probabilities)
- Define the detailed test event as D
- Conditional probabilities:

  $P(D|S) = 0.6$ $P(D|U) = 0.2$
- Goal is to find $P(S|D)$

Bayes' Theorem Example

(continued)

Apply Bayes' Theorem:

$$P(S|D) = \frac{P(D|S)P(S)}{P(D|S)P(S) + P(D|U)P(U)}$$

$$= \frac{(0.6)(0.4)}{(0.6)(0.4) + (0.2)(0.6)}$$

$$= \frac{0.24}{0.24 + 0.12} = 0.667$$

So the revised probability of success, given that this well has been scheduled for a detailed test, is 0.667

Bayes' Theorem Example

(continued)

- Given the detailed test, the revised probability of a successful well has risen to 0.667 from the original estimate of 0.4

Event	Prior Prob.	Conditional Prob.	Joint Prob.	Revised Prob.
S (successful)	0.4	P(D|S)=0.6	(0.4)(0.6) = 0.24	0.24/0.36 = 0.667
U (unsuccessful)	0.6	P(D|U)=0.2	(0.6)(0.2) = 0.12	0.12/0.36 = 0.333

Sum = 0.36

Counting Rules

- Rules for counting the number of possible outcomes

- Counting Rule 1:
  - If any one of k different mutually exclusive and collectively exhaustive events can occur on each of n trials, the number of possible outcomes is equal to

$$\boxed{k^n}$$

Counting Rules

(continued)

- Counting Rule 2:
  - If there are k_1 events on the first trial, k_2 events on the second trial, … and k_n events on the nth trial, the number of possible outcomes is

$$\boxed{(k_1)(k_2)\ldots(k_n)}$$

  - Example:
    - You want to go to a park, eat at a restaurant, and see a movie. There are 3 parks, 4 restaurants, and 6 movie choices. How many different possible combinations are there?
    - Answer: (3)(4)(6) = 72 different possibilities

Counting Rules

(continued)

- Counting Rule 3:
  - The number of ways that n items can be arranged in order is

$$n! = (n)(n - 1)\ldots(1)$$

  - Example:
    - Your restaurant has five menu choices for lunch. How many ways can you order them on your menu?
    - Answer: $5! = (5)(4)(3)(2)(1) = 120$ different possibilities

Counting Rules

(continued)

- Counting Rule 4:
  - Permutations: The number of ways of arranging X objects selected from n objects in order is

$$\boxed{{}_nP_x = \frac{n!}{(n-X)!}}$$

  - Example:
    - Your restaurant has five menu choices, and three are selected for daily specials. How many different ways can the specials menu be ordered?
    - Answer: $nPx = \frac{n!}{(n-X)!} = \frac{5!}{(5-3)!} = \frac{120}{2} = 60$ different possibilities

Counting Rules

(continued)

- Counting Rule 5:
  - Combinations: The number of ways of selecting X objects from n objects, irrespective of order, is

$$\boxed{{}_nC_x = \frac{n!}{X!(n-X)!}}$$

  - Example:
    - Your restaurant has five menu choices, and three are selected for daily specials. How many different special combinations are there, ignoring the order in which they are selected?
    - Answer: ${}_nC_x = \frac{n!}{X!(n-X)!} = \frac{5!}{3!(5-3)!} = \frac{120}{(6)(2)} = 10$ different possibilities

Chapter Summary

- Discussed basic probability concepts
  - Sample spaces and events, contingency tables, simple probability, and joint probability
- Examined basic probability rules
  - General addition rule, addition rule for mutually exclusive events, rule for collectively exhaustive events
- Defined conditional probability
  - Statistical independence, marginal probability, decision trees, and the multiplication rule
- Discussed Bayes' theorem
- Examined counting rules